

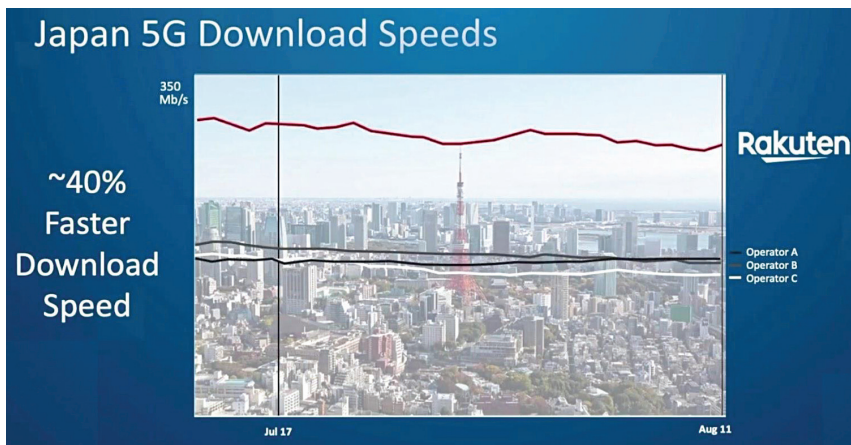


Fig. 3: 5G download speed in downtown Tokyo

they are to deliver an optimised implementation that captures the essence and can really unlock the shift from fixed-function systems to software-defined systems.

Necessity of data-centre platform for specific 5G implementations

The 5G base station data centre switches to factory control systems and to old system retail outlets built from fixed-function hardware that are being replaced by systems which enable developers to program. The various functions and workloads are being lifted up and provide liberty to software developers to define them. In this way, software developers are able to continuously evolve, improve, and tailor systems to gain more control, flexibility, and security in their networks and edge systems in edge computing.

The industry is racing to analyse, process, and store data at the edge, where those new services are being delivered. This could be edge IoT applications running on a customer's premise or in a Kono facility, or networking workloads running in a telecom data centre or an internet exchange point.

5G delivers flexibility via network functions running in the cloud, and has many ways to deliver multiple options for deployment

models that range from public to private 5G enterprises. This explores deployment options to access connectivity and deliver use cases with a level of control and investment that they choose for themselves.

For example, public and private networks can work together to deliver connectivity directly to end users. Consider a smart sports stadium. The owners of the stadium can operate an on-premise network for dedicated connectivity and control the operations of the stadium. At the same time, public networks provide connectivity to the fans inside the stadium.

Enterprises can own a really high-capacity mobile network for themselves. The reliable one has lower latency and can support time-sensitive applications. Enterprises across whole different industries look to take advantage of new technologies like AI/machine learning, advanced robotics, augmented reality, and virtual reality. 5G private networks will allow enterprises to develop and deploy new applications, including real-time control and time-sensitive low-latency applications.

The market opportunity has two main components. One is the legacy fixed-function network market, which we expect to hold steady as we shift the market to run on server based solutions. The other is the net-

work and edge infrastructure, where we expect to see enormous growth across cloud data centres, core networks, and edge computing segments.

Transition from fixed-function to fully-programmable

The programmable GPU devices have replaced the huge fixed-function graphics cards that were used at workstations with sophisticated SoCs. These new software based implementations enable the development and deployment of new algorithms in the field and a variety of new capabilities in software by developers.

Platforms are being transformed and horizontalised, and software is being disaggregated at the system level. Previously, big cloud providers used to build their own switches based on merchants. For example, switch a6 was added to Linux, and it replaced each box as per applications.

Today, all of the top 20 cloud service providers build their own networking equipment and write the software themselves. This reduces cost, gives them much more control, and they also have an army of software engineers to do it. More than half of all the core network workloads have already moved from proprietary fixed-function boxes into virtual workloads running on standard servers.

This has also started steadily in the radio access network (RAN), with the transition from ASIC based round boxes to virtual RAN or vRAN running in software equipment, such as Ericsson and Nokia moving from fixed silicon to CPUs running workloads in software. All early service provider VRAN deployments are running on Xeon and Flex RAN.

The uptake of VRAN is expected to grow from a relatively small percentage in 2021 to approaching a quarter of all deployments by 2025. At the same time, a lot of network software is moving to cloud native. A lot of people questioned whether you could really build a 5G network